The electromagnetic spectrum

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate electromagnetic wavelength, frequency and propagation.

SUCCESS CRITERIA - I can...

1. Use c=fλ in vacuum.
2. Order spectral regions by wavelength.
3. Distinguish electromagnetic from mechanical waves.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A closed-open pipe has length 0.3 m. Use λ=4 L for the fundamental and v=340 m/s. Find its fundamental frequency.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $c = f\lambda$; $c = 3.0\times10^{8}$ m/s.
Electromagnetic waves in vacuum. Frequency identifies spectral region; mechanical sound needs a medium.

Worked reasoning: In vacuum, all electromagnetic waves travel at $c=3\times10^{8}$ m/s. A frequency of 10^{8} Hz gives wavelength 3 m. Higher frequency means shorter wavelength at the same speed.

Your turn: Draw two waves with a common speed but different wavelengths; label which has higher frequency.



Marzano check - circle: ① ② ③

03 Read and interpret the model

Order radio, visible and X-rays from longest to shortest wavelength 0. Which has greatest frequency in vacuum?

Radio	IR	Visible	UV	X-ray

Label increasing frequency and wavelength.

Marzano check - circle: ① ② ③

04 Apply the physics

Compare two signals in vacuum: light of wavelength 600 nm and radio of frequency 100 MHz. Use $c = 3 \times 10^{8}$ m/s, 1 nm = 10^{-9} m and 1 MHz = 10^{6} Hz. Find the light's frequency and radio wavelength. Complete the table.

Signal	Frequency (Hz)	Wavelength (m)
Light	____	600×10^{-9}
Radio	100×10^{6}	____

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

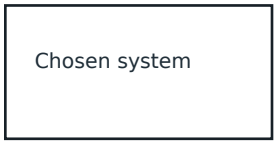
Why can sunlight cross space but ordinary sound from a speaker cannot cross a vacuum?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A heat engine operates between 600 K and 400 K and absorbs 800 J. Find maximum ideal work and explain why actual work is usually lower.
Label energy entering, leaving and stored.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3

07 Exit check

A wave has $f = 2.4 \times 10^9$ Hz in vacuum. Find λ. Identify the spectral region.

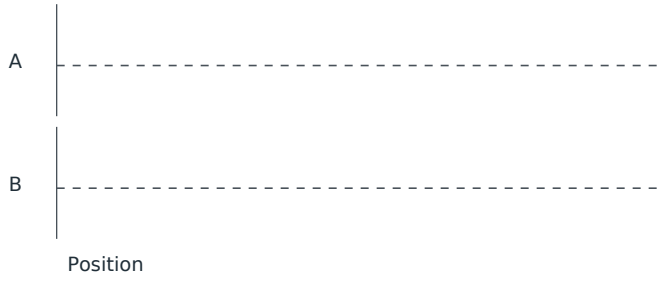
Radio	IR	Visible	UV	X-ray

Label increasing frequency and wavelength.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Visible light has wavelength 600 nm. Find frequency using $1\ \text{nm} = 10^{-9}$ m. Compare with light of wavelength 300 nm.



Marzano check - circle: 1 2 3